



Urban parks, value uplift and green gentrification: An application of the spatial hedonic model in the city of Brisbane

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ABSTRACT

This paper explores the impact of urban parks on real estate prices making use of a hedonic price approach. Focusing on Brisbane, Australia, as a case study site, we use spatial hedonic models to analyse housing sales data across 15,000 sales transactions to investigate the effects of parks on nearby housing prices, paying attention to park typology and classification. Our findings indicate that recreational and sport parks are differently associated with price variations. The study also examines a specific and significant inner-city park currently undergoing a major redevelopment—namely Victoria Park. Our analysis of the Victoria Park site seeks to quantify the value uplift, that is, the future increase in property prices as a result of the transformation of the current private golf course in this location into a new publicly accessible parkland. This study's property economics modelling analysis indicates the conversion of Victoria Park from a golf course to public parkland will increase property prices by an average of 3 % for properties located within 750 m of the park. The article concludes with a discussion of value capture opportunities that these findings present as well as challenges of green gentrification for this and similar urban renewal projects and possible policy responses.

1. Introduction

Urban green space can refer to different typologies of areas, characterized by various functions and degrees of naturalness, such as urban parks and gardens, outdoor sports areas, school gardens, grassland and wetlands. Irrespective of their characterization, urban green areas provide a variety of benefits to the population, which range from the opportunity to spend time in nature and enjoy aesthetically pleasing landscapes to environmental benefits such as improvement in air quality, water run-off regulation, conservation of biodiversity, and so on (Chiesura, 2004). More generally, it has been recognized that urban green space offers a wide range of ecosystem services, including biophysical, environmental, cultural, and perceptive dimensions (Andersson et al., 2015; Bolund and Hunhammar, 1999; Wiggering et al., 2006). Additionally, many economic benefits are associated with urban green areas and, more generally, with environmental goods and services (Ambrey and Fleming, 2014; Bertram and Rehman, 2015). It is well known that the use and transformation of environmental systems have

been shown to have a variety of economic effects and repercussions, to the extent that the environment is considered a key (positive or negative) externality within the urban economics literature (Palmquist, 2005). In light of the growing importance of environmental factors in urban and land economics, specific evaluation tools have been developed that allow the value of the environment and urban green spaces to be estimated in order to establish and assess the foreseeable benefits of certain actions involving their use and transformation (van der Heide and Heijman, 2013).

The majority of studies on economic valuation of urban green spaces employ a Total Economic Value (TEV) approach (Pearce and Turner, 1990; Harnik, 2014). The TEV approach is based on the assumption that any good or service is composed of various attributes, some of which are concrete and easily measured, while others may be more difficult to quantify. The overall value of the good or service is given by the sum of all of these components. Different evaluation techniques are available for estimating the TEV. According to Pearce and Turner (1990), the methods can be divided into two main families: i) Revealed preferences

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Table 1
Hedonic pricing applications regarding the impact of urban parks on real-estate market.

Author	Title	Journal	HPM results
Dell'Anna et al (2022)	Urban Green infrastructures: How much did they affect property prices in Singapore?	Urban Forestry & Urban Greening	The analysis was conducted using OLS, GWR, SAR, SARAR, and SEM. Apartment values increase differently according to the type of UGIs (urban parks, regional parks, and nature areas).
Cheung and Fernandez (2021)	Close enough: housing price effects of urban parks, reserves and volcanic parks in Auckland, New Zealand	International Journal of Housing Markets and Analysis	Using OLS, the results shows that the logged distance to the nearest urban park is insignificant related to the housing values.
Piaggio (2021)	The value of public urban green spaces: Measuring the effects of proximity to and size of urban green spaces on housing market values in San José, Costa Rica	Land Use Policy	Increasing proximity to metropolitan parks and size of neighborhood parks increases property values.
Zhang et al (2021)	The rising and heterogeneous demand for urban green space by Chinese urban residents: Evidence from Beijing	Journal of Cleaner Production	The willingness to pay for parks has continued to grow during the study period in Beijing, but there exists heterogeneous demand related to the housing of residents and different types of parks.
Nur et al (2020)	Assessing the economic value of urban green spaces in Kuala Lumpur	Environmental Science and Pollution Research	OLS results show that decreasing the distance between the public urban parks and house locations is unambiguously associated with increasing the house price. GWR captured that the significant value with the expected sign of each environmental attributes variables is located at different sub-districts.
Yuan et al (2020)	Amenity effects of urban facilities on housing prices in China: Accessibility, scarcity, and urban spaces	Cities	The coefficient of D_PARK (Distance along road network to nearest park (m)) is significantly negative in most of the considered cities whereas the coefficient of S_PARK (Number of parks within 1 km from the community (n)) is significantly positive, indicating that surrounding green amenities can increase the values of real estates.
Fernandez and Bucaram (2019)	The changing face of environmental amenities: Heterogeneity across housing submarkets and time	Land Use Policy	Different effects on residential property transactions in Auckland (New Zeland) can be detectable according to various parks features (e.g., ownership and maintenance)
Kim et al (2019)	Understanding the local impact of urban park plans and park typology on housing price: A case study of the Busan metropolitan region, Korea	Landscape and Urban Planning	Urban park characteristics showed slightly different results across the categories: completed parks (an increase of more than 6 % in housing price), theme park (an increase of more than 3 % in housing price), small park and children's park (more than 2 % and 0.4 %, respectively).
Łaskiewicz et al. (2019)	Can proximity to urban green spaces be considered a luxury? Classifying a non-tradable good with the use of hedonic pricing method	Ecological Economics	Proximity does not result in increased apartment prices for all parks and forests. Proximity to selected parks rises in consecutive apartment price sub-segments.
Li et al (2019)	Analyzing housing prices in Shanghai with open data: Amenity, accessibility and urban structure	Cities	The results suggest that accessibility to public parks have positive effects on housing prices in Shanghai.
Panduro et al (2018)	Eliciting preferences for urban parks	Regional Science and Urban Economics	Two different measures of park are analysed. The implicit price of one hectare of park within a 1000 m radius is 53.25 EUR per ha per year for the average apartment corresponding to an increase in annual rent of 0.33 % per additional ha. For reducing distance to the nearest park by a meter, the price is 0.59 EUR per meter per year, corresponding to an increase in annual rent of 0.03% per meter.
Iqbal and Wilhelmsson (2018)	Park proximity, crime and apartment prices	International Journal of Housing Markets and Analysis	The proximity of parks as an environmental amenity has an effect on apartment prices and the impact is different over the residential market segments in Stockholm. Moreover, various types of parks may differ in their impact, for instance, grass parks and park blocks are more desirable in Stockholm than landscape parks and neighbourhood parks. The effects of crimes in parks influence apartment prices negatively.
Vom Hofe et al (2018)	Are homeowners willing to pay more for access to parks? Evidence from a spatial hedonic study of the Cincinnati, Ohio, USA park system	Journal of Regional Analysis and Policy	Decreasing the distance to a park by one meter appreciates of the value of the average house by \$3.44 to \$6.27. A one hectare increase in the park size appreciates house values by \$22.24 to \$22.48 and the presence of park facilities for recreational activities depreciates house values by an estimated \$4612 to 5300 for properties within a 500 m radius to the nearest park.
Engström and Gren (2017)	Capturing the value of green space in urban parks in a sustainable urban planning and design context: Pros and cons of hedonic pricing	Ecology and Society	Parks have a positive effect on property value and this effect tends to increase with reduced distance to the parks.
Jun and Kim (2017)	Measuring the effect of greenbelt proximity on apartment rents in Seoul	Cities	Authors examine how the implicit value of the amenity by Seoul's greenbelt (GB) is reflected in apartment rents in Seoul. The marginal implicit value of decreasing the distance to the GB by one kilometer yields about a \$34 drop in monthly rent.
Park et al (2017)	Park accessibility impacts housing prices in Seoul	Sustainability	The results show that when distance from the park increases, the value of the park inherent in the housing price decreases. Moreover, the greater walking accessibility, to the park, the higher the park value inherent in housing prices.
Schläpfer et al (2015)	Valuation of landscape amenities: A hedonic pricing analysis of housing rents in urban, suburban and periurban Switzerland	Landscape and Urban Planning	The examination of the value of landscape amenities in different community types reveals some consistent influences of income and urban densities on the role of amenities and disamenities in housing markets.
Brander and Koetse (2011)	The value of urban open space: Meta-analyses of contingent valuation and hedonic pricing results	Journal of Environmental Management	Percentage change in house prices due to a 10 m decrease in distance to open space. The unweighted average effect is 0.137 %, i.e., house prices increase when located closer to open space.

methods, that derive the value of an environmental and landscape resource from the observation of individuals' decisions in real markets. These methods include Travel Cost Method (Voke et al., 2013) and Hedonic Pricing method (Freeman III et al., 2014); ii) Stated preferences methods, that are based on the creation of a simulated market data collection by asking individuals for their Willingness To Pay (WTP) for using a certain good or service, or Willingness To Accept (WTA) abandoning a certain good. Contingent Valuation Method (Carson, 2000; Price, 2000) and Conjoint Analysis/Choice Experiments method belong to this family.

This study builds the established TEV methodological framework and extends it by applying a hedonic pricing methodology to explore the effect of urban parks on real estate prices in the city of Brisbane, Queensland, Australia. In recent years, different studies have highlighted that the presence of urban parks positively affects the price of apartments in the surrounding areas. In particular, the use of the Hedonic Modelling approach allows researchers to estimate the value of environmental amenities by appraising the role of proximity to urban parks in the determination of the price of real estate (Morancho, 2003; Jim and Chen, 2006). While it has been demonstrated that proximity to urban green areas plays an important role in the determination of housing prices in surrounding areas, the impact of the specific characteristics of urban parks—including their physical features, spatial context, and typology—has been less explored (Kim et al. 2019; Czembrowski and Kronenberg, 2016; Holt and Borsuk, 2020). In order to fill this gap, the present paper investigates how proximity to urban parks affects housing prices in the city of Brisbane in terms of increase in property values, focusing in particular on the value uplift associated with the different park characteristics and typologies. Beyond the benefits delivered by urban parks, the article also discusses the negative social impacts that these areas produce in terms of gentrification, in that the added value to housing created by urban greening interventions leads to higher rental prices, and thus can result in the displacement of economically vulnerable residents (Anguelovski et al., 2018).

Starting from this premise, the objective of the article is twofold. First, it aims to provide an exploration of the effect of urban parks on housing prices in the city of Brisbane, according to the different park typologies. Second, drawing on the results of the empirical model and considering the transformation of a private 18-hole golf course into a publicly accessible parkland, the paper quantifies the potential impact on the surrounding real estate properties, thus providing the first estimation of the value uplift associated with the development of this urban park. The impact of the presence of the new park in terms of increase in housing prices is also discussed from the point of view of possible green gentrification effects that could negatively impact the area under examination.

The study will be of interest to urban greening researchers, property values, and decision makers involved in the planning and management of urban green renewal projects, as its findings (a) demonstrate an approach for quantifying both the value uplift and associated urban ecosystem services provided by urban parks and (b) contribute to debates surrounding the issue of green gentrification.

2. Literature review

The importance of quantifying the value of non-market goods can be explained by the increasing demands on natural capital and ecosystems services on our urbanized planet (Guo et al., 2010) and the need for people to live in an inclusive, healthy, resilient, safe and sustainable environment (Loh et al., 2020; Yigitcanlar et al., 2019; Bottero et al., 2020; Caprioli et al., 2021). Many scholars point to the role and benefit of natural capital to improve human well-being, social cohesion, and social inclusion, both qualitatively and quantitatively (Bertram and Rehdanz, 2015; Dennis and James, 2016; Ekins, 2003; Jim and Chen, 2009). A special attention has been devoted in the literature to the analysis of natural capital in exploring the relationship between urban green spaces

and property valuation (Jones et al., 2019; Valente et al., 2020; Ekins, 2003). In particular, the Hedonic Price Method – HPM (Rosen, 1974) has been widely applied to estimate the contribution to property values delivered by environmental amenities (Crompton, 2001). Indeed, since Ridker and Henning (1967) found that hedonic pricing models are statistically significant in the context of environmental economics, increasing attention has been given to the evaluation of externalities generated by natural resources and environmental amenities and their effect on house prices (Geoghegan, Wainger, and Bockstael, 1997; McLeod, 1984). In recent years, the availability of urban data and advances in geo-statistics have made valuation studies of urban green spaces more and more important (Holt and Borsuk, 2020).

The rationale behind HPM is that the house is a composite good and its value is determined by multiple characteristics, including tangible housing elements (e.g., surface, floor, state of conservation, etc.) and intangible elements (e.g., landscape quality, environmental services, etc.). Accordingly, the hedonic function allows a marginal price of each characteristic to be assigned, leading to a monetary estimation of each housing attribute.

An analysis of the literature shows the ongoing interest in the adoption of this approach for estimating the value of environmental externalities (Sirmans et al., 2006). Recent publications analyse, among other factors, the spillover effects related to ecological lands and landscape proximity (Xiao et al., 2019), waterfronts (Dahal et al., 2019), urban rivers (Chen et al., 2019), forests (Lamhamedi et al., 2021), the flood risks perception (Sado-Inamura and Fukushima, 2019), environmental amenities across space (Hiebert and Allen, 2019), preference between detention and retention ponds (Sohn et al., 2020), and the impact of multimodal accessibility to green space (Chen et al., 2021).

With specific reference to the value of urban green areas, Kong et al. (2007) pointed out that for the city of Jinan (China) a 1 % improvement in accessibility to parks will lead to an increase in housing price per square meter of 1.6 %. Analogously, for the city of Salo (Finland), Tyrvaenen and Miettinen (2000) found that individuals are willing to pay 4.9 % more for houses with a forest view. An analysis of Turin (Italy) illustrates how the quality of the area, including proximity to green areas, affects in a positive way the value of a property (D'Acci, 2014). Dell'Anna et al. (2022) expressly analysed the impact of green infrastructures on housing prices in the city of Singapore, pointing out the existence of an influence of city parks, regional parks, and nature areas on apartment values. Specific studies by Kim et al., 2019, Nur et al., 2020 and Piaggio, 2021 in the cities of Seoul, Kuala Lumpur and San José (Costa Rica), respectively, explore the role of size and characteristics of the urban green areas in the determination of the prices. Cheung and Fernandez (2021) examine the different effects of three categories of green spaces (i.e. urban parks, regional reserves and volcanic parks) on residential property transactions in Auckland (New Zealand). Several studies that analyse the effects of urban greening, including greenbelts, urban tree programs and creation of parks, on property values, indicate that buyers appreciate the importance of living in neighbourhoods with greener and denser vegetation (Jun and Kim, 2017; Li et al., 2019; Payton et al., 2008; Yuan et al., 2020). In another study focused on the city of Stockholm, the negative influence of crimes in parks on apartment prices is estimated (Iqbal and Wilhelmsson, 2018).

With specific reference to urban parks, Table 1 details the most recent applications that investigate the application of HPM for estimating the spillovers effects of this typology of areas.

As far as the city of Brisbane is concerned, the analysis of the literature highlights a number of applications of HPM for estimating the effects of different urban goods and services, including transport infrastructures, urban forestry and flood control. In particular, Table 2 details the main fields of research of these studies with a focus on the characteristic variables used for the implementation of the different HPM approaches. Our literature review substantiates the novelty of the research described in this article fourfold: (1) It is the first local application of an HPM valuation on the effects of urban parks on property

Table 2
Hedonic pricing studies the city of Brisbane.

Author	Field of research	Characteristic Variables
Siripanich et al. (2019)	Public transport accessibility	- housing variables (e.g., number of bedrooms), - neighbourhood's variables (e.g., percentage of people with weekly income greater than \$1500)- public transit accessibility indicators (e.g., proportion of transfer trips of all originated trips)
Rajapaksa et al. (2017)	Flood dynamics	- housing variables (e.g., number of bedrooms), - locational variables (e.g., direct distance in meters to the closest park)- flood related variables (distance to the river, flood-affected)- elevation variables (e.g., elevation of low-income suburbs (elev ² low-income suburbs))
Bryant (2017)	Infrastructure charges	- housing variables (e.g., number of bedrooms), - locational variables (e.g., relative Socio-economic Advantage and Disadvantage (SEIFA)), - jurisdictional variables (e.g., percentage rate of change in population (LGA)), - policy variables (e.g., percentage annual developer charge adopted per lot, lagged by one year)
Plant et al. (2016)	Street Tree Cover Targets	- housing variables (e.g., number of bedrooms), - locational (e.g., distance to arterial road (km)), - suburb variables (e.g., suburb housing age (years)), - tree canopy cover variables (e.g., percentage tree cover within 30 m of property)
Rajapaksa et al. (2017)	Flood dynamics	- housing variables (e.g., number of bedrooms), - socio-economics characteristics (e.g., median income, access to transport)- locational variables (e.g., direct distance in meters to the closest park)- flood related variables (distance to the river, flood-affected)
Mulley et al. (2016)	Proximity to transport infrastructure	- housing variables (e.g., number of bedrooms), - locational (e.g. street network distance to the nearest bus stations (100's meters)),
Rajapaksa et al. (2016)	Flood dynamics	- housing variables (e.g., number of bedrooms), - locational variables (e.g., direct distance in meters to the closest park)- sales variables (e.g., dummy variable for the year of sale: 1 if sale occurs in 2006, 0 otherwise)
Rambaldi et al. (2013)	Inundation-prone Suburb	- housing variables (e.g., number of bedrooms)

values in the city of Brisbane. (2) An additional element of innovation of this study is related to the fact that the characteristics of the parks are considered in the estimation of the effects on housing pricing. (3) The paper's originality is further corroborated by the study's development of the evaluation following the approach of the spatial hedonic models, an implementation of the traditional HPM that is receiving more and more attention in the real estate appraisal (LeSage and Pace, 2009; Krause and Bitter, 2012; Bottero et al., 2018). (4) Finally, the effects of urban parks in terms of increases in property values are also discussed with regards to the possible impacts of green gentrification. Our review suggests that the studies that examine green gentrification impacts grounded in the empirical results of hedonic modelling are very limited at the moment (Jo Black and Richards, 2020; Bockarjova et al., 2020).

3. Methodology

3.1. Hedonic price modelling

The Hedonic Pricing Method (HPM) is based on the idea that real estate properties are not homogeneous goods (Rosen, 1974). Their market value is influenced by the presence of a bundle of attributes:

locational, structural, temporal, geographical, and environmental. Each characteristic has an implicit price embodied in the selling price; the implicit price is revealed only from observed values – revealed preferences – of differentiated products with a specific quantity of each attribute. The theory is based on the concept that housing prices are divided between housing characteristics (intrinsic) and location characteristics (extrinsic). More formally, the basic model can be described by Eq. (1):

$$p = \beta_0 + \sum \beta_k S_k + \sum \beta_j L_j + u \quad (1)$$

Where p is the vector of housing prices, S_k is a matrix of intrinsic structural characteristics, L_j is a matrix of extrinsic location characteristics, β_0 is the intercept coefficient, β_k , β_j , are the corresponding parameters and u is a vector of random errors.

For empirical estimates, HPM relies on a regression technique, which has been criticised for causing a series of problems that can lead to biased estimates, such as functional form specification, spatial heterogeneity, spatial autocorrelation, housing quality change, multicollinearity, and heteroscedasticity (Elhorst, 2014). In response to these critiques, HPM has been continuously evolving, with the help of more powerful computation methods and improved systems. One of these are Geographic Information Systems (GIS), which allow for more precise identification and valuation from a spatial perspective (Anselin, 1988). Indeed, recent research has paid particular attention to the issue of spatial dependence (spatial autocorrelation) and spatial structure (spatial heterogeneity) in regression models (Anselin, 1988; Paelinck and Klaasen, 1979; Dubin, 1992; Tsai et al., 2017). Specific models have thus been developed in order to deal with spatial issues for overcoming possible biases in the regression estimates.

As far as spatial heterogeneity is concerned (Wang and Chen, 2020), it is possible to state that real estate market is influenced by elements that are not uniformly distributed in the urban space. To address this issue, many researchers proposed the use of local regression models, such as GWR (Geographically Weighted Regression) (Oshan et al, 2019; Massimo et al., 2019). GWR is a spatial statistical technique able to capture the heterogeneity characterizing the area under investigation, developing a set of local linear models based on subsets of observations and providing findings on the geographical scale of the estimated effects.

The GWR model provides an estimate of the position characteristics for each observation in the model, as well as a single bandwidth parameter that allow to investigate the geographic scale of the effects.

The GWR model is specified as follows:

$$y = \beta_0 + \sum_{k=1}^p \beta_{ik} x_{ik} + e_i, i : 1, 2, \dots, n \quad (2)$$

Where, for location i , y_i is the dependent variable, β_0 is the intercept coefficient, x_{ik} is the k -th variable, β_{ik} is the local regression coefficient of the k -th variable and e represents the random error. Location i is represented by geographic coordinates (u_i , v_i).

GWR estimator for local parameter estimates is expressed as follows:

$$\hat{\beta}(i) = [X'W(i)X]^{-1}X'W(i)y \quad (3)$$

where X is n by k matrix of explanatory variables, $W(i)$ is the n by n diagonal matrix of weights that considers each observation with reference to its distance from the position of location i , $\hat{\beta}(i)$ is a vector of coefficients and y is a vector of observations of the dependent variable.

As far as spatial dependence is considered, spatial lag models are routinely used in real estate evaluations where house prices are at least partly determined by the prices of the neighbourhood properties (or Spatial Autoregressive model – SAR). The spatial log model addresses the potential spatial interaction between the dependent variable (house price) and its neighbours by introducing the hedonic function through a specific price weight matrix as follows:

$$p = \beta_0 + \rho Wp + \sum \beta_k s_k + \sum \beta_j L_j + \varepsilon \quad (4)$$

$$\varepsilon = \rho W\varepsilon + u \quad (5)$$

where the term ρWp corresponds to the weighted average price of the neighbouring observations, the parameter ρ is known as the autocorrelation coefficient and the term u is the uncorrelated error term. It is important to note that W represents the spatial weights matrix. This $N \times N$ matrix, being N the size of the dataset, is composed by the elements w_{ij} that are the spatial weights, with w_{ij} non-zero when i and j are hypothesised to be neighbours and 0 otherwise.

The SAR model also implies that there exist direct spillover effects between the prices of neighboring properties (LeSage and Pace, 2009). For the marginal implicit prices calculation, following Won Kim et al. (2003), the formula (6) can be used:

$$\frac{\delta p_i}{\delta x_{ij}} = \beta_j (I - \rho Wp)^{-1} \quad (6)$$

According to this formula it is possible to state that the house price in the location j is not only affected by a marginal change of one characteristic of the property, but also by the marginal changes of the neighbors. The total impact of a change in one characteristic is thus given by the sum of direct and indirect impacts, allowing a sort of price adjustment between neighboring properties.

Another model used in spatial regression is the Spatial Error Model (SEM), which takes into account the fact that errors are spatially correlated due to unobserved variables or measurement errors. The SEM model can be represented by Eqs. (7) and (8):

$$p = \beta_0 + \sum \beta_k s_k + \sum \beta_j L_j + \varepsilon \quad (7)$$

$$\varepsilon = \lambda W\varepsilon + u \quad (8)$$

where W is the spatial weights matrix, λ is the spatial error coefficient and u is the uncorrelated error term.

It is important to notice that the formula (4) does not apply to the SEM, where the error correction makes up for the omission of variables related to externalities and local public goods and implicit marginal prices are constant, supposing the functional form is linear.

In this study we used two types of spatial econometric models, in addition to the traditional hedonic price model (OLS – Ordinary Least Square), to address spatial autocorrelation: 1) SAR, 2) SEM. Details about the application of these models are provided in Section 4.

3.2. The Study Area

Brisbane is a subtropical city located in Queensland, Australia (Fig. 1). It is the third-largest city in Australia with a population of around 1.25 million in the Brisbane City Council Local Government Area (LGA) and 2.5 million in the Greater Capital City Statistical Area for Brisbane, as defined by the Australian Bureau of Statistics (ABS). Brisbane city comprises over 150 suburbs and localities and covers an area of 1.3 sq km. Brisbane City Council is the largest local government in Australia with 26 wards and 27 councillor positions. There is currently significant sprawling development on the outskirts of the city, but the majority of the population is concentrated in a strip through the centre. This is where formal and informal creative and commercial activities overlap (Mengi et al., 2020). Being situated in the subtropics, Brisbane is a city with considerable amounts of green space per capita, and public green spaces and parks are distributed fairly evenly across the city.

The present study is focused on the Inner Brisbane area (Fig. 1), a homogeneous area characterised by a similar density compared to other external areas. The area covers a surface of around 1367 sq km for an overall population of 250,207 inhabitants (Australian Bureau of Statistics, 2016). As for the public parks being considered, the city has a wide variety of parks, ranging from small pocket parks to large district parks. Fig. 2 shows the distribution of urban parks and the apartment sales data

used in the analysis in the area under consideration.

3.3. Urban green space classification

According to Brisbane City Council (2014) and, in particular, secondary data sources related to public parks and land for community facilities network, there are seven types of park in the city of Brisbane and most of them are subcategorised into other specific types. Appendix 1 contains a description of all park types, and specifies, where applicable, the size, the users and an example image. However, it was out of the scope of this study to consider all park types specified within the Brisbane City Council (2014) documentation on parks. There are two reasons for this limitation in scope. Firstly, the spatial open dataset contained on the Brisbane City Council website¹ only contains information about the location of each park in the city, without any references to these typologies. As a result, this would have required us to manually define the specific subcategories for each green area. Secondly, some subcategories can be considered equivalent to more general ones and some functions can be contained in many types of parks, in particular, in those more general, such as *Sports Parks* or *Recreational Parks*.

For these reasons, our analysis instead considered a previous classification scheme found in the Brisbane Priority Infrastructure Plan of 2011. This document classifies the urban parks into two main categories: *Sports Parks* and *Informal Recreation Parks*, which are both subcategorised based on the park size and location in the city. Starting from the open dataset with the park functions and assets,² we thus have defined the main use of each park dividing them in informal recreation or sports parks and taking into account whether they are local or district/metro-sized parks.

3.4. Data and variables

The sale prices of 15,330 single residential property sales between 2016 and 2019 were identified within the Inner Brisbane study area with also details about numbers of bedrooms, bathrooms, and garage space house attributes, available from the CoreLogic³ RP Data set. Moreover, accessibility variables were calculated using GIS by measuring the Euclidean distances to the nearest public park, considering the different typologies of sites. Table 3 presents a summary of the variables used in the analysis. It is worth mentioning that following Wen et al. (2015) and Sohn et al. (2020), in the first version of the model the full range of variables, including location, neighbouring, environmental and structural variables were included. However, as the Inner Brisbane area considered in the present study is a quite homogeneous area described by similar features from the point of view of presence of sport and educational facilities, commercial areas and transport routes, the location and neighbouring variables resulted to be not significant being the same for most of the sample. Therefore, by a backward elimination processes the current set of variables, which is composed of structural variables and environmental variables, has been identified. This approach is aligned with other studies in the context of hedonic models, which do not include location or neighbouring variables in the model because these variables do not vary as much within each area as to be included in the evaluation (Anthon et al., 2005; Mulley et al. 2016; Rajapaksa et al., 2016; Rambaldi et al., 2013).

4. Results

This section presents the study's results and findings in two parts.

¹ <https://www.data.brisbane.qld.gov.au/data/dataset>

² <https://www.data.brisbane.qld.gov.au/data/dataset/park-facilities-and-assets>

³ <https://www.corelogic.com.au/>

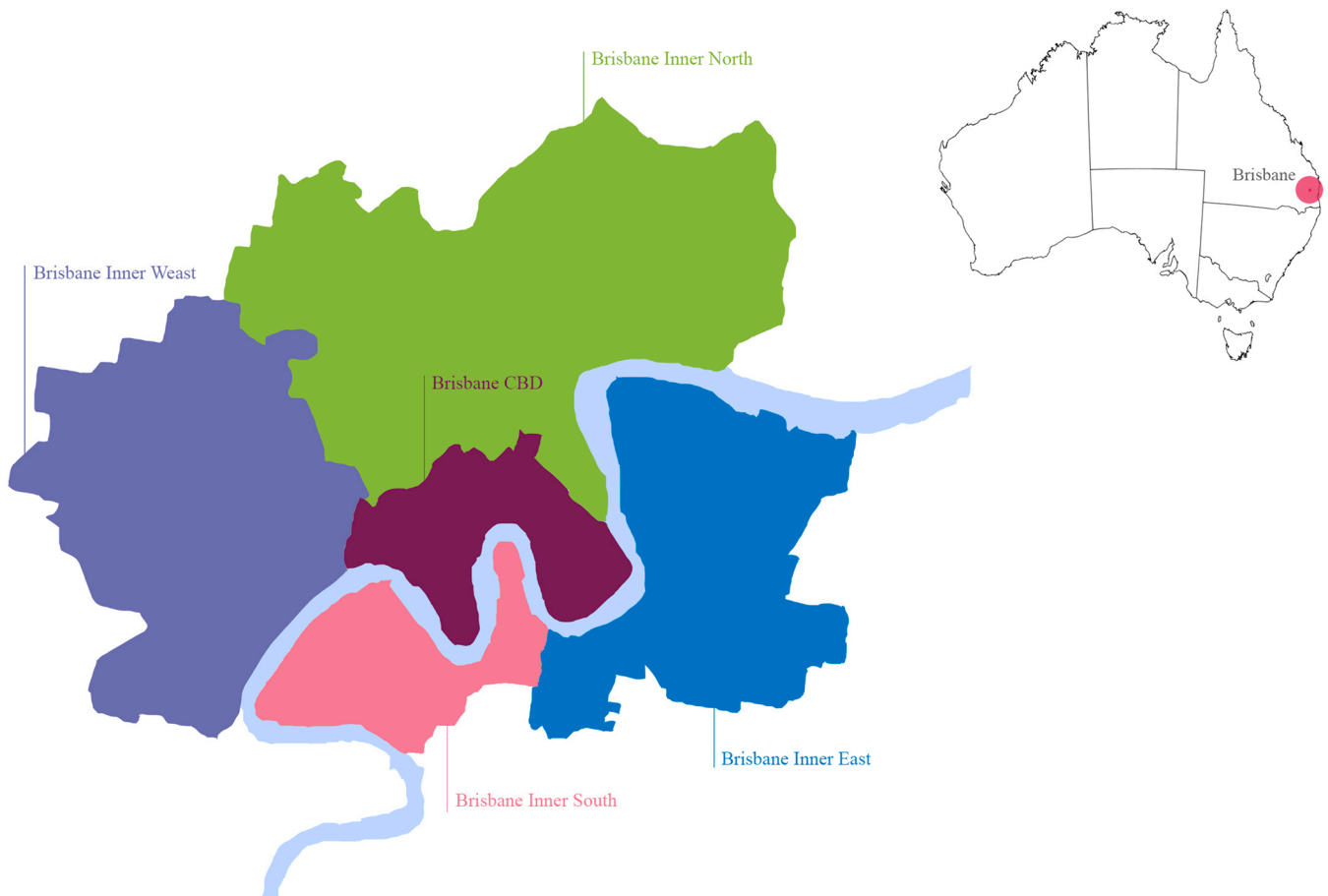


Fig. 1. On the right, the location of Brisbane in the Queensland State (Australia) and, on the left, a map of the Brisbane Inner City suburbs.

First, we discuss the analysis of our data looking at the relationship between proximity recreational vs sports parks. We find a 7 % increase in value for recreational parks and 1 % decrease for sports parks. Second, we examine the specific data pertaining to the Victoria Park redevelopment and its impact on value uplift. Here, we calculate a 3 % increase in value.

4.1. Proximity to recreational vs sports parks

4.1.1. Ordinary least squares models

According to the study methodology described in Section 2, several models have been tested including Ordinary Least Square (OLS) model, Spatial Error Model (SEM), and Spatial Autocorrelation Model (SAM). Following the approach taken by most hedonic models in the literature (Rosen, 1974), this study uses the natural log of price as dependent variable in the multiple regression function, as represented in Eq. (6):

$$\ln(P) = \beta_0 + \beta_1 \text{BED} + \beta_2 \text{BATH} + \beta_3 \text{CAR} + \beta_4 \text{PARK} + \varepsilon \quad (9)$$

In this model, P is the vector of the transaction price of apartments; BED, BATH and CAR are the number of bedrooms, bathrooms and car parking spaces, respectively; PARK is a Boolean variable, which assumes the value 1 if the apartment is located within 750 m from a park and the value 0 otherwise. It has to be noted that according to the design standards for urban park spaces in the city of Brisbane (Brisbane City Council, 2014), 750 m corresponds to the walking distance to urban parks. Indeed, the desired accessibility standard for parks is up to 750 m walking distance, meaning that all residents should be able to walk 750 m or less to access a park. The variable PARK has been further detailed in the different versions of the model, in order to estimate the role of the different park typologies on housing prices. More in details,

following the Municipal Plan of the city of Brisbane, two typologies of urban parks have been considered, recreational parks and sport parks, that were further specified according to their extension, leading to analyse four types of parks: small recreational parks (surface less than 0,5 ha), large recreational parks (surface greater than 0,5 ha), small sport parks (surface less than 1 hectare) and large sport parks (surface greater than 1 hectare).

The symbols β_1 , β_2 , and β_3 represent the regression coefficients for the corresponding terms, whereas ε represents the random error. The error is calculated as the difference between the observed and predicted value and represents all of the variations in the dependent variable not explained by the considered independent variables.

In the development of the estimation, the OLS model was tested first and the results are reported in Table 4. As it is possible to see, most of the housing characteristic variables significantly affect housing price.

4.1.2. Local regression model

In this research we applied the GWR model to determine the relationships between the transaction prices and urban parks across the city. Indeed, the GWR model is being used for spatial prediction rather than for inference (Harris et al., 2010). In particular, the MGWR software was used (Oshan et al., 2019) and a GWR model was applied that uses an adaptive bisquared spatial kernel. A Golden Section was used to define the bandwidth searching that optimizes (i.e., minimizes) the AICc (Corrected Akaike Information Criterion). For calibrating GWR models, we adopted a Gaussian model (Tables 5 and 6).

Fig. 3 shows the local parameter estimates for urban parks under investigation derived from the GWR model.

For the urban parks (Fig. 3a) the GWR results show that the coefficients are uniformly negative over almost the entire territory,



Fig. 2. Distribution of urban parks and the apartment sales data used in the analysis area.

Table 3

Variables used in the analysis.

Variables		Unit	Min	Max	Mean	Standard Deviation
Apartment characteristics	Price	[AUS \$]	1099	2365,0000	840,893,355	740,052,483
	Bedroom	[number]	0	20	2643	1370
	Bathroom	[number]	0	16	1711	0896
	Car	[number]	0	17	1462	1019
Distance	Urban park	[meter]	3262	5789,337	282,086	168,731
	Recreational urban park	[meter]	3262	5789,337	310,515	186,042
	Sport urban park	[meter]	8841	6627,113	636,947	344,772

confirming the results of the OLS model. Indeed, the model proved that the lower the distance from the urban parks, the higher the housing prices. Moreover, the GWR calibration allowed to highlight the spatial variation of recreational urban parks influence. As shown in Fig. 3b, recreational parks have a great influence on the West region due to the presence of the Mt Coot-tha Forest. Analogously, Fig. 3c shows positive features in the North-Central area where sport parks are limited or absent.

The values of the local coefficients are roughly all negative, indicating a positive effect determined by the proximity to the parks of this category (Fig. 3b). Therefore, the GWR model was able to validate the signs of the coefficients of the global model, confirming the positive impact of recreational parks and the negative influence of sport parks on housing prices.

4.1.3. Spatial autoregressive models

In addition to the OLS model, two types of spatial regression models

were developed to control spatial autocorrelation: the spatial autoregressive (SAR) model and the spatial error model (SEM). In particular, the software GeoDa was employed (Anselin et al., 2006). GeoDa addresses the problem of spatial dependence through distance weight matrices, calculating contiguity weights. In the present application, the Queen's contiguity method was used to create the weight matrix W assuming that all adjacent houses affect the value of a given house (Sohn et al. 2020). The Queen criterion determines neighbouring units as those that have any point in common, including both common boundaries and common corners. Accordingly, the weight of the surrounding houses assumed the value of 1 if they shared a border or vertex of a given house, and 0 if not.

After the contiguity weight matrix calculation, the Moran's Index (MI) was computed to measure the spatial autocorrelation effect. In detail, the MI was calculated by relating the natural logarithm of the total price ($\ln$ Price) of each transaction and the average value of the neighbouring ones (lagged $\ln$ Price) (Fig. 4). The results confirmed that

Table 4

Results of the OLS model considering the different typologies of urban parks in Brisbane.

REGRESSION RESULTS – LINEAR MODEL (OLS) – NUMBER OF OBSERVATIONS = 15,334						
	All parks	Recreational/ Sport Parks	Local Recreational / District Recreational Parks	Local Sport / District Local/District Sport Parks	Local/District Recreational Parks and Local / District Sport Parks	Sport Parks and Local / District Recreational Parks
Variables	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE
<i>Constant</i>	12.650 *** (0.034)	12.628 *** (0.030)	12.686 *** (0.020)	12.733 *** (0.012)	12.692 *** (0.020)	12.694 *** (0.020)
<i>Bed</i>	0.211 *** (0.005)	0.211 *** (0.005)	0.2106 *** (0.005)	0.210 *** (0.005)	0.210 *** (0.005)	0.211 *** (0.005)
<i>Bath</i>	0.026 *** (0.007)	0.025 *** (0.007)	0.0266 *** (0.007)	0.026 *** (0.007)	0.026 *** (0.007)	0.026 *** (0.007)
<i>Car</i>	0.076 *** (0.005)	0.076 *** (0.005)	0.0766 *** (0.005)	0.076 *** (0.005)	0.076 *** (0.005)	0.076 *** (0.005)
<i>Urban parks</i>	0.068 ** (0.033)					
<i>Recreational parks</i>		0.105 *** (0.029)				
<i>Sport parks</i>		-0.021 ** (0.010)				-0.018 * (0.014)
<i>Recreation parks < 0.5 ha</i>			-0.003 (0.014)		0.0026 (0.014)	-0.0002 (0.014)
<i>> 0.5 ha</i>			0.037 *** (0.014)		0.045 *** (0.014)	0.040 *** (0.009)
<i>Sport parks < 1.0 ha</i>				-0.045 *** (0.013)	-0.049 *** (0.013)	
<i>> 1.0 ha</i>				-0.013 (0.010)	-0.015 (0.010)	
S.E. of regression	0.535	0.535	0.535	0.535	0.535	0.535
R Squared	0.308	0.309	0.308	0.309	0.309	0.308
Adjusted R- squared	0.308	0.308	0.308	0.309	0.309	0.308

*** indicates probability $p < 0.01$; ** indicates probability $p < 0.05$; * indicates probability $p < 0.1$.**Table 5**

OLS and GWR regression results for urban parks (Bandwidth used: 95.000, p-value < 0.05).

	OLS	GWR				
	Coeff.	Mean	STD	Min	Median	Max
Intercept	-0000	0.146	2.815	-5.780	-0.018	59.015
Bed	0.452	0.438	0.349	-3.874	0.424	5.196
Bath	0.035	0.004	0.270	-4.761	0.026	2.663
Car	0.121	0.094	0.190	-1.151	0.086	1.300
Urban parks	-0.031	0.036	2.330	-5.682	-0.035	45.991
Adj. R ²	0.0309	0.593				
AICc	37865.221	32009.718				

the autocorrelation phenomenon occurred in the area under investigation as the MI which shows a discrete level of spatial autocorrelation.

The four quadrants in the graph provide a classification of four types

of spatial autocorrelation: high-high, low-low, high-low and low-high. Spatial autocorrelation analysis can also be visualised in a specific cluster map where the high-high, low-low, high-low and low-high clusters are shown (Fig. 5). This map is based on the so-called LISA (Local Indicators of Spatial Association) that allows to delineate hot spots and cold spots in the real estate market that represent specific submarkets of housing prices. More specifically, the LISA map indicates a significant spatial clustering of high and low prices in the city of Brisbane; moreover, the map clearly shows segmented housing markets between inner city and external areas. The map shows that most of the high price clusters are concentrated in the area of East Brisbane with New Farm being the most expensive suburb in the city.

Subsequently, the SAR and SEM models have been developed and the results of the estimations are reported in Tables 7 and 8, respectively. Following Anselin (2008), specific tests have been developed to determine the choice of models, including Log-Likelihood value, Akaike criterion (AIC) and Schwarz criterion (SC). According to the literature, the

Table 6

OLS and GWR regression results for green and sport parks (Bandwidth used: 95.000, p-value < 0.05).

	OLS	GWR				
	Coeff.	Mean	STD	Min	Median	Max
Intercept	-0000	0.023	0.725	-15.075	-0.008	10.500
Bed	0.451	0.430	0.341	-5.695	0.424	2.655
Bath	0.036	0.012	0.262	-1.706	0.024	2.083
Car	0.120	0.097	0.201	-1.052	0.085	1.645
Recreational parks	-0.027	-0.057	0.649	-8.721	-0.013	9.899
Sport parks	0.027	-0.003	0.626	-13.803	0.013	21.046
Adj. R ²	0.0309	0.593				
AICc	37861.250	32009.718				

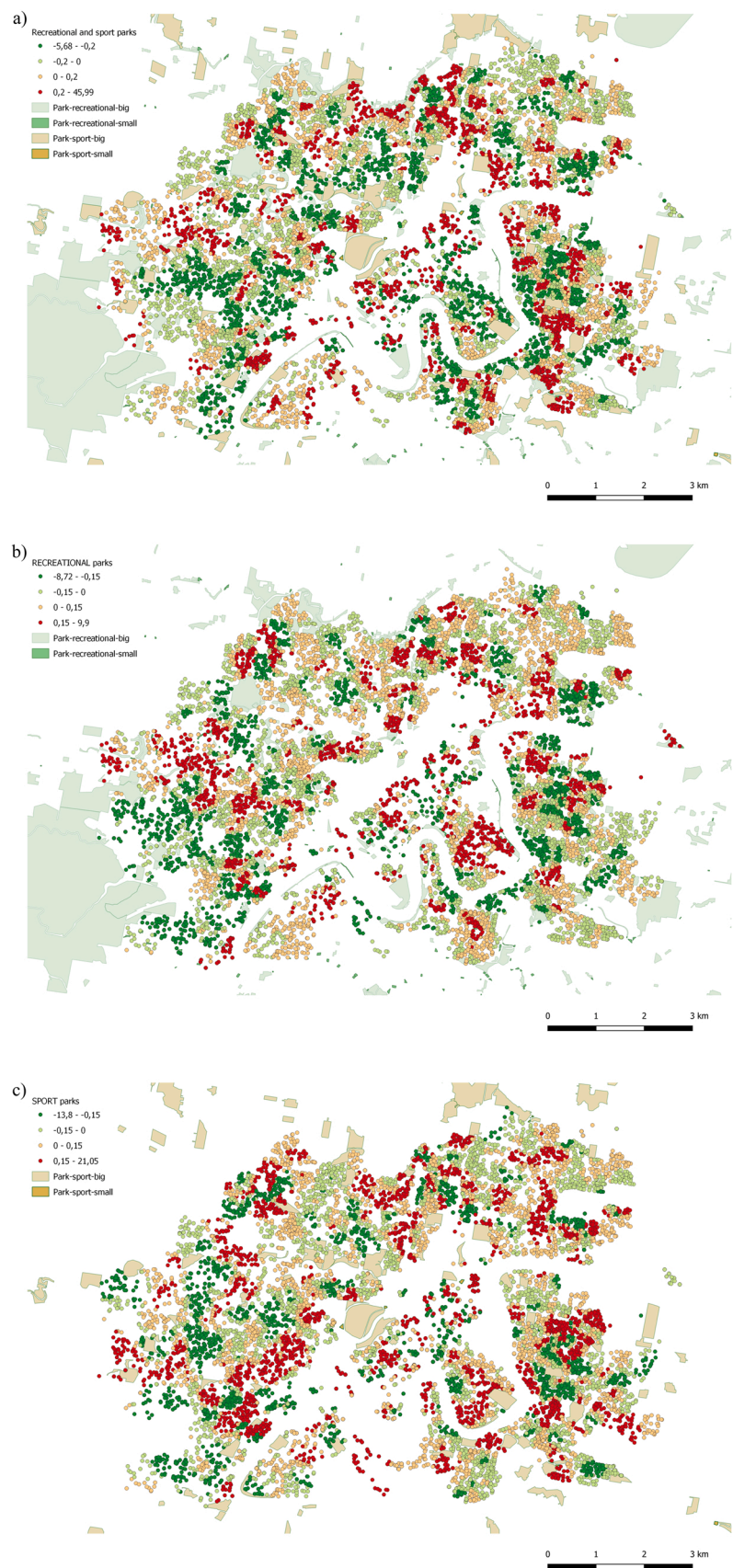


Fig. 3. Local GWR estimates for the urban parks in Brisbane.

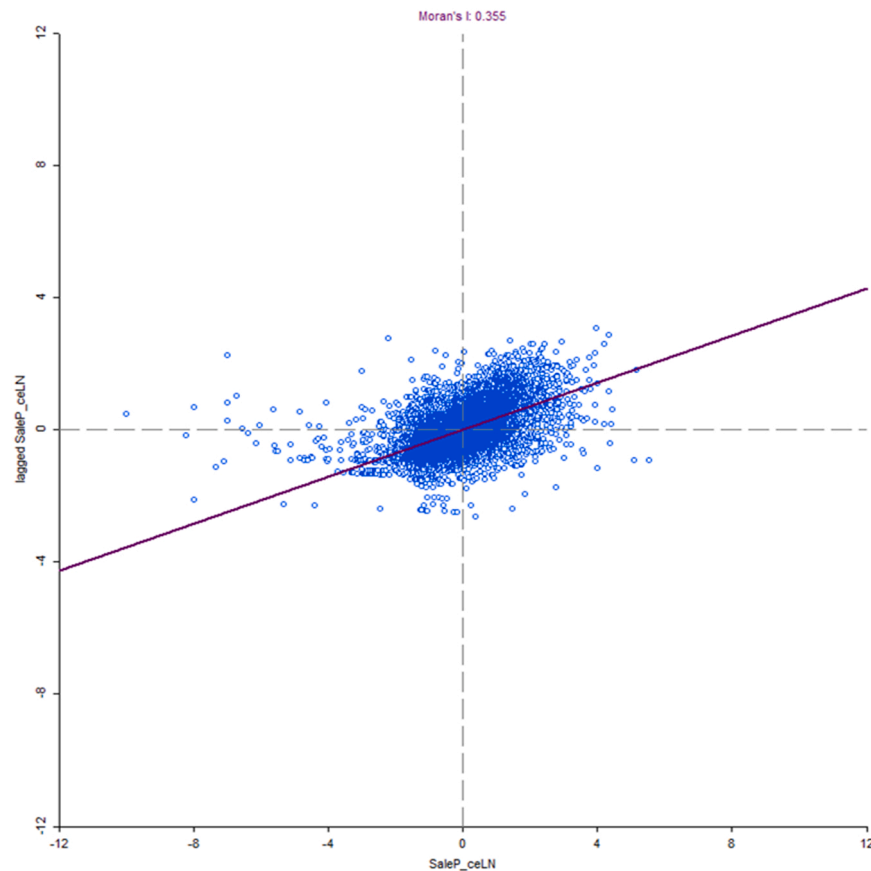


Fig. 4. Moran Index.

best performing spatial model has the Log-Likelihood value closer to zero, and the lowest values of AIC and SC. Moreover, also the Lagrange Multiplier (LM) statistical index was calculated in order to further verify the results of the model.

The results indicate that all the variables related to the property intrinsic characteristics have a significant relationship with the price. As expected, these characteristics are positively correlated to the price, meaning that the higher the number of bathroom/bedroom/car parking, the higher the price. As far as the urban parks are considered, the results of the model show that their role is slightly different across the considered categories. For urban parks in general, the results show that the presence of a park contributes to a value uplift of around 4–6 % in property value (second column of Tables 4, 7, 8). An in-depth analysis taking into account the characteristics of the urban park revealed that recreational parks are positively associated with the housing prices, whereas sports parks are negatively associated (third column of Tables 4, 7, 8). In particular, we found that proximity to recreational parks correlates with a predicted increase of 7 % in housing prices whereas, by comparison, the presence of a sports park translates into a 1 % decrease. This opposite effect could be related to the real or perceived negative impact that sports parks can have on the surrounding neighbourhoods in relation to noise, traffic and other negative impacts that might be present. Unexpectedly, when considering the dimension of the urban parks, the findings of the model did not show any divergence between large and small parks. Both showed an expected increase of 3 % in the housing prices (fourth column of Tables 4, 7, 8). By comparison, for sports parks, the differences in the dimension were not appreciated by the market and their effects turned out to be not significant according to the performed analysis.

4.2. Value uplift from the Victoria Park redevelopment

As described above, the results of the evaluation model allowed us to understand how the presence of urban parks affects the housing prices in the city of Brisbane. Our findings indicate that quantifying the impact of new urban parks on residential property prices in terms of creation of externalities can lead to useful insights. Expanding on previous work on land value analysis and rapid scenario planning (RAISE⁴ project) (Pettit et al., 2020), the study experimented with using the outputs of the hedonic model to rapidly estimate the changes in property values resulting from the planned redevelopment and future construction of a new park in the city of Brisbane.

Victoria Park Golf Course is a privately operated 18-hole golf course, located across the Brisbane suburbs of Kelvin Grove and Herston. In 2019 Brisbane City Council released plans to redevelop the area into an expansive new public park. Redevelopment plans and concepts are currently being considered as part of a community consultation process,⁵ but as of June 2021, the private golf course was closed and repurposed as public parkland during the redevelopment process.⁶ Fig. 6 shows the vision of the park as emerging from the consultation process, imagining a natural retreat, an urban park for adventure, discovery and reconnection.

Victoria Park is one of Brisbane's largest and oldest green spaces. The park is located on the city centre's northern edge, surrounded by residential suburbs, business precincts and large institutions including universities, hospitals, schools and churches. The surrounding suburbs

⁴ <https://research.qut.edu.au/designlab/projects/raise/>

⁵ <https://www.brisbane.qld.gov.au/victoriapark>

⁶ <https://www.abc.net.au/news/2021-05-26/victoria-park-golf-course-open-to-the-public-from-july/100168208>

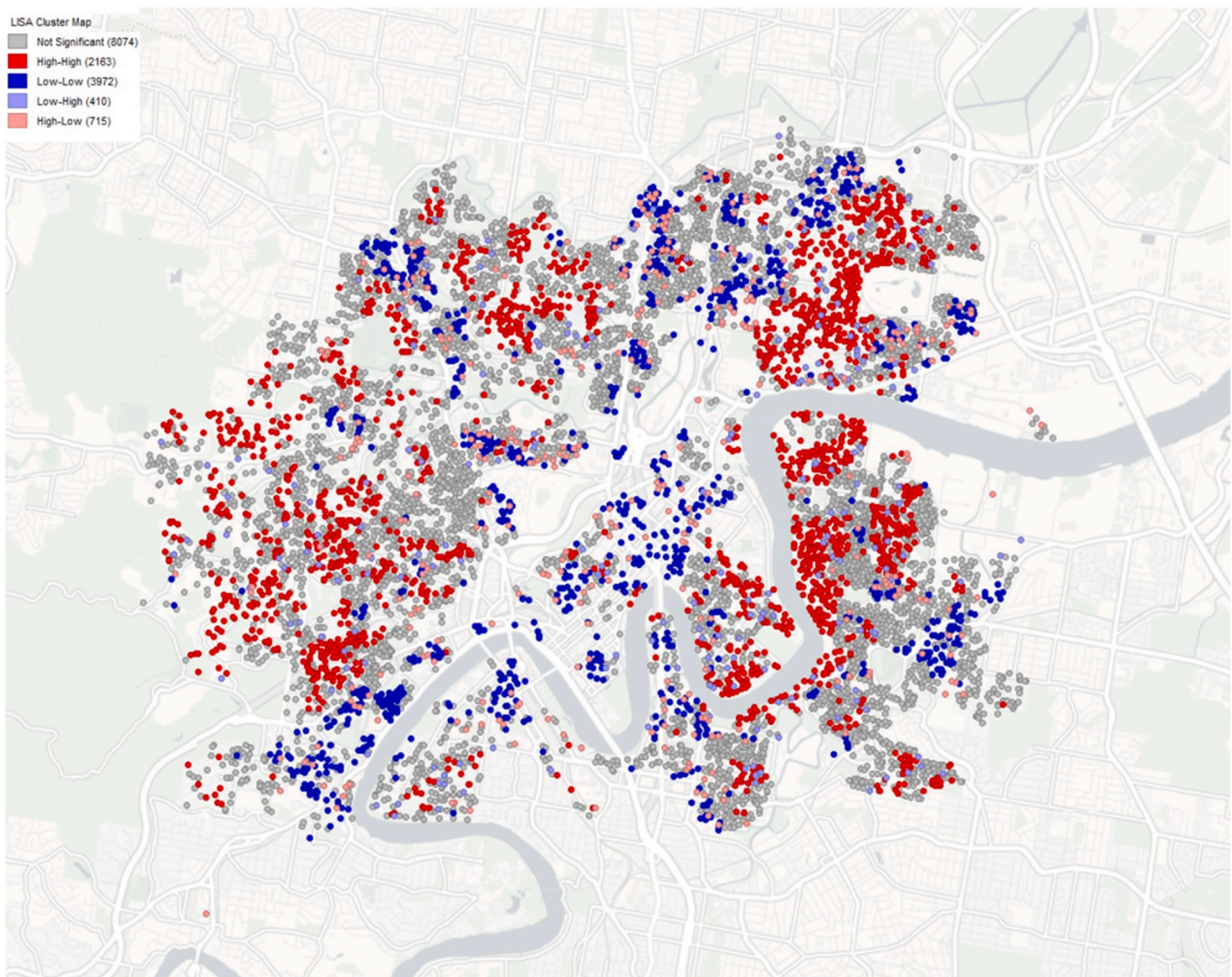


Fig. 5. Cluster map of hot spots and cold spots in Brisbane real estate market.

are some of the most densely populated and economically diverse areas of Brisbane. The resident population of these localities is projected to increase from approximately 52,500 people in 2019–80,500 people by 2041. As part of the Victoria Park Vision, Council is committed to protecting and enhancing almost 45 ha of this greenspace for future generations.

To the purpose of the evaluation, Fig. 7 shows the Victoria Park Golf Course and the distribution of apartment sales data used in the analysis within the considered area. Based on the results of the hedonic modelling, we estimate that the transformation of the golf course into a recreational green park would lead to a 3 % average increase of housing prices for assets located within a 750 m radius of the park. If these findings were applied to the new Victoria Park development the predicted average value uplift for the area would be estimated at around 63 million AU\$ (Table 9).

5. Discussion

While in this section we discuss the study's specific local case of Brisbane, it is worth mentioning upfront that the research has knowledge transfer potential and applicability beyond the geographic focus of Brisbane as the impact of urban park on property values and the issue of green gentrification are not limited to Brisbane or Australia, but apply to many cities internationally.

According to the general results of our model, we can state that in the city of Brisbane recreational natural parks and sports parks have different effects in terms of increase in housing prices. Following our findings, proximity to natural parks is perceived as valuable by individuals who are willing to pay more for a home close to a park, whereas proximity to sports parks is not considered as a positive externality in the real estate market and their presence leads to a slight decrease in housing prices. It can be stated that this negative spillover might be due to negative impacts associated with sport parks such as traffic congestion, lack of parking or noise pollution. Surprisingly, the differences in the size of the parks were not perceived as relevant in the evaluation model, which shows the same impact for large and small parks.

As far as the case of the new Victoria Park is considered, based on our research, we were able to estimate the value uplift related to the proposed redevelopment of the golf course into a public parkland in the city of Brisbane. Modelled findings indicate a potential overall value uplift of around AUD 63 million, based on the increase of house prices in the area surrounding the park.

It is likely that the development of Victoria Park will result in increased economic activity and revitalisation of local businesses, such as cafés, restaurants, and retailers. This and the attractiveness of the new park will likely drive property price increases as the desirability of the area increases. Our study's property economics research suggests an

Table 7

Results of the SAM model considering the different typologies of urban parks in Brisbane.

REGRESSION RESULTS – SPATIAL LAG MODEL (SAM)– NUMBER OF OBSERVATIONS = 15,334						
	All parks	Recreational/ Sport Parks	Local Recreational / District Recreational Parks	Local Sport / District Sport Parks	Local/District Recreational Parks and Local / District Sport Parks	Sport Parks and Local / District Recreational Parks
Variables	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE
<i>W_Sale Price</i>	0.508 *** (0.010)	0.508 *** (0.010)	0.509 *** (0.010)	0.507 *** (0.010)	0.508 *** (0.010)	0.509 *** (0.010)
<i>Constant</i>	6.025 *** (0.131)	6.009 *** (0.130)	5.994 *** (0.130)	6.085 *** (0.127)	6.012 *** (0.130)	6.001 *** (0.130)
<i>Bed</i>	0.140 *** (0.005)	0.141 *** (0.005)	0.14 *** (0.005)	0.140 *** (0.005)	0.141 *** (0.005)	0.141 *** (0.005)
<i>Bath</i>	0.048 * ** (0.007)	0.047 *** (0.007)	0.048 *** (0.007)	0.048 *** (0.007)	0.048 *** (0.007)	0.048 *** (0.007)
<i>Car</i>	0.063 *** (0.005)	0.063 *** (0.005)	0.063 *** (0.005)	0.063 *** (0.005)	0.063 *** (0.005)	0.063 *** (0.005)
<i>Urban parks</i>	0.043 (0.030)					
<i>Recreational parks</i>		0.077 *** (0.027)				
<i>Sport parks</i>		-0.020 ** (0.009)				0.034 *** (0.013)
<i>Recreation parks < 0.5 ha</i>			0.031 ** (0.013)		0.034 *** (0.013)	0.033 *** (0.012)
<i>> 0.5 ha</i>			0.031 ** (0.012)		0.035 * (0.012)	-0.020 ** (0.009)
<i>Sport parks < 1.0 ha</i>				-0.013 (0.012)	-0.018 (0.012)	
<i>> 1.0 ha</i>				-0.013 (0.009)	-0.017 * (0.009)	
<i>LN_PRICE (mean)</i>	13.429	13.429	13.429	13.429	13.429	13.429
<i>R Squared</i>	0.432	0.432	0.432	0.432	0.433	0.433
<i>Akaike criterion</i>	21783.2	21775.5	21773.6	21782.6	21770.1	21770.1
<i>Schwarz criterion</i>	21829	21828.9	21827.1	21836.1	21838.8	21831.2
<i>Log Likelihood</i>	-10885.6	-10880.7	-10879.8	-10884.3	-10876	-10877
<i>S.E. of regression</i>	0.485	0.484	0.484	0.485	0.484	0.484
<i>Lag coefficient (Rho)</i>	0.508	0.508	0.510	0.507	0.508	0.509
<i>LM (lag)</i>	4033.884***	4034.743***	4057.174***	4014.007***	4008.491***	4062.729***
<i>Robust LM (lag)</i>	525.397***	539.921***	532.241***	516.867***	536.948***	537.183***

*** indicates probability $p < 0.01$; ** indicates probability $p < 0.05$; * indicates probability $p < 0.1$.

average increase in property prices in the vicinity of 3 %. This result casts doubts over urban green space strategies that may have conflicting outcomes. While this correlation can be considered positive from the perspective of adjacent property, land and businesses owners, new park developments can also lead to gentrification, altering housing opportunities and thus negatively impacting lower-income communities (Kabisch and Haase, 2014; Krueger and Gibbs, 2007; Wolch et al., 2014; Basu and Nagendra, 2021). This paradoxical effect has been called *green gentrification*, that is the transformation of—and capitalization upon—an environmental “bad” when turned into an environmental “good” (Gould and Lewis, 2012). In the Brisbane case, furthermore, the transformation of the golf course into a park can be ascribed to an environment-friendly agenda of the public actor in which there is not actually an “environmental bad” but rather a “neutral,” since the golf course is largely perceived as an already green area, albeit for a specific use, and in which the narrative is less on ecological aspects and more on the creation of a world-class park, thus strengthening the city branding strategy as a global “new world” city and the related economic development strategy.

The resulting uneven ethical space between environmental awareness, governance in cities, and concerns of inequality (Quastel, 2009) cannot be simply overlooked when dealing with the obvious, powerful,

message of the coming back into the public realm of a formerly privatized area. There are, in fact, at least two aspects that would be interesting to consider to have a fuller picture of this otherwise clever transformation. The first aspect would be to consider that the transformation of the golf course into a public park is not a stand-alone project but is happening in an area in which other projects of even greater magnitude are being planned or implemented (such as Brisbane Live, Cross River Rail, Brisbane Metro Line 1, Herston Quarter, etc.). A second aspect might consider the possible effects of such a gentrification process in terms of displacement of former inhabitants and the implementation of selective mechanisms for future ones, in an area where the marketable qualities are not those of a typical deprived neighbourhood or of a brownfield area, rather real estate investors may be attracted to capitalize on already existing improvements (Gould and Lewis, 2017; Angotti, 2008). The broader impacts of a green gentrification process should then be considered and planned for with appropriate policy and regulation (Pearsall and Eller, 2020). The potential increase in property values may exacerbate housing affordability problems, and force members of the local community to leave, but this is not the only problem we may foresee for future developments within the park area and in a larger part of the inner city.

Table 8

Results of the SEM model considering the different typologies of urban parks in Brisbane.

REGRESSION RESULTS – SPATIAL ERROR MODEL (SEM) – NUMBER OF OBSERVATIONS = 15,334						
	All parks	Recreational/ Sport Parks	Local Recreational / District Recreational Parks	Local Sport / District Sport Parks	Local/District Recreational Parks and Local / District Sport Parks	Sport Parks and Local / District Recreational Parks
Variables	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE	Coefficient (β) with SE
<i>Constant</i>	12.886 *** (0.057)	12.839 *** (0.034)	12.815 *** (0.024)	12.854 *** (0.015)	12.817 *** (0.025)	12.825 *** (0.025)
<i>Bed</i>	0.168 *** (0.005)	0.168 *** (0.005)	0.168 *** (0.005)	0.168 *** (0.005)	0.168 *** (0.005)	0.168 *** (0.005)
<i>Bath</i>	0.027 *** (0.007)	0.027 *** (0.007)	0.027 *** (0.007)	0.027 *** (0.007)	0.027 *** (0.007)	0.027 *** (0.007)
<i>Car</i>	0.066 *** (0.0047)	0.066 *** (0.005)	0.066 *** (0.005)	0.066 *** (0.005)	0.066 *** (0.005)	0.066 *** (0.005)
<i>Urban parks</i>	-0.039 (0.057)					
<i>Recreational parks</i>	–	0.020 (0.032)				
<i>Sport parks</i>	–	-0.016 (0.011)				0.016 (0.017)
<i>Recreation parks < 0.5 ha</i>	–		0.016 (0.017)		0.018 (0.017)	0.022 (0.015)
<i>> 0.5 ha</i>	–		0.021 (0.015)		0.025 (0.015)	-0.016 (0.011)
<i>Sport parks < 1.0 ha</i>	–			-0.031 ** (0.015)	-0.034 ** (0.015)	
<i>> 1.0 ha</i>	–			-0.003 (0.011)	-0.004 (0.011)	
<i>LAMBDA</i>	0.594 *** (0.011)	0.593 *** (0.011)	0.594 *** (0.011)	0.593 *** (0.011)	0.593 *** (0.011)	0.594 *** (0.011)
<i>LN_PRICE (mean)</i>	13.429	13.429	13.429	13.429	13.429	13.429
<i>R Squared</i>	0.434	0.434	0.435	0.434	0.435	0.435
<i>Akaike criterion</i>	21922.7	21923	21922.1	21920.6	21920.6	21922
<i>Schwarz criterion</i>	21960.8	21968.8	21967.9	21966.5	21981.7	21975.5
<i>Log Likelihood</i>	-10956.330	-10955.477	-10955.058	-10954.315	-10952.321	-10954.021
<i>S.E. of regression</i>	0.483	0.483	0.483	0.483	0.483	0.483
<i>Lag coefficient (Lambda)</i>	0.594	0.593	0.594	0.593	0.593	0.594
<i>LM (error)</i>	3937.743 ***	3907.118 ***	3942.073 ***	3930.653 ***	3880.150 ***	3937.442 ***
<i>Robust LM (error)</i>	429.257 ***	412.295 ***	417.141 ***	433.514 ***	408.607 ***	411.896 ***

*** indicates probability $p < 0.01$; ** indicates probability $p < 0.05$; * indicates probability $p < 0.1$.

We believe a broader discussion about these impacts is necessary to develop appropriate protections for members of the local community who may be susceptible to these types of impacts. In particular, precautionary policies that directly impact on rent gap accumulation should be considered (Immergluck and Balan, 2018), considering the usually low provision of public affordable housing in future developments. Similarly, taxation mechanisms should be considered to help existing homeowners to keep their properties, avoiding an immediate capitalization of the increase in their property values, and to help tenants to continue to be able to afford their rents. The same could be said for small businesses and existing shops, which may or may not be impacted by the possibility of planning new large commercial facilities. Precautionary policies may be based on precautionary tales, as in the case of many North American cases (Atlanta, Chicago, New York, Vancouver, etc.) where green gentrification researchers have mostly focused their attention in the past (Dale and Newman, 2009; Quastel, 2009; Immergluck and Balan, 2018; Rigolon and Németh, 2018). Other precautionary tales may build on (continental) European cases, where it is more common for public actors to play a more direct role of public actors at the national or municipal level, following a tradition of welfare state provision and intervention through regeneration policies (see, for instance, Anguelovski et al., 2018). This same concept of regeneration

could, however, be seen as another way of mainstreaming gentrification processes, whether critically discussed (Smith, 1996) or charged with promises of future improvements (Florida, 2002).

6. Conclusion

In this study, we applied the hedonic pricing method to explore the potential value of urban parks in the city of Brisbane, considering different typologies of urban parks, namely recreational and sports parks, and for each category, and examining the relationship between housing prices and a set of variables describing the houses and the neighborhood characteristics, including the distance to urban parks. According to our findings, there is a proximity effect between urban parks and the increase in housing prices (Kim et al., 2019) with differences across the park typologies.

The capability of HPM to measure the value of environmental goods and services is widely accepted in the literature; our study has shown the possibility of improving the estimations by considering the specific features of the amenities under investigation. Moreover, the research has experimented with how the results of the evaluation could be used to support the decision-making process in the context of urban planning, considering both the positive impacts of urban green areas in terms of

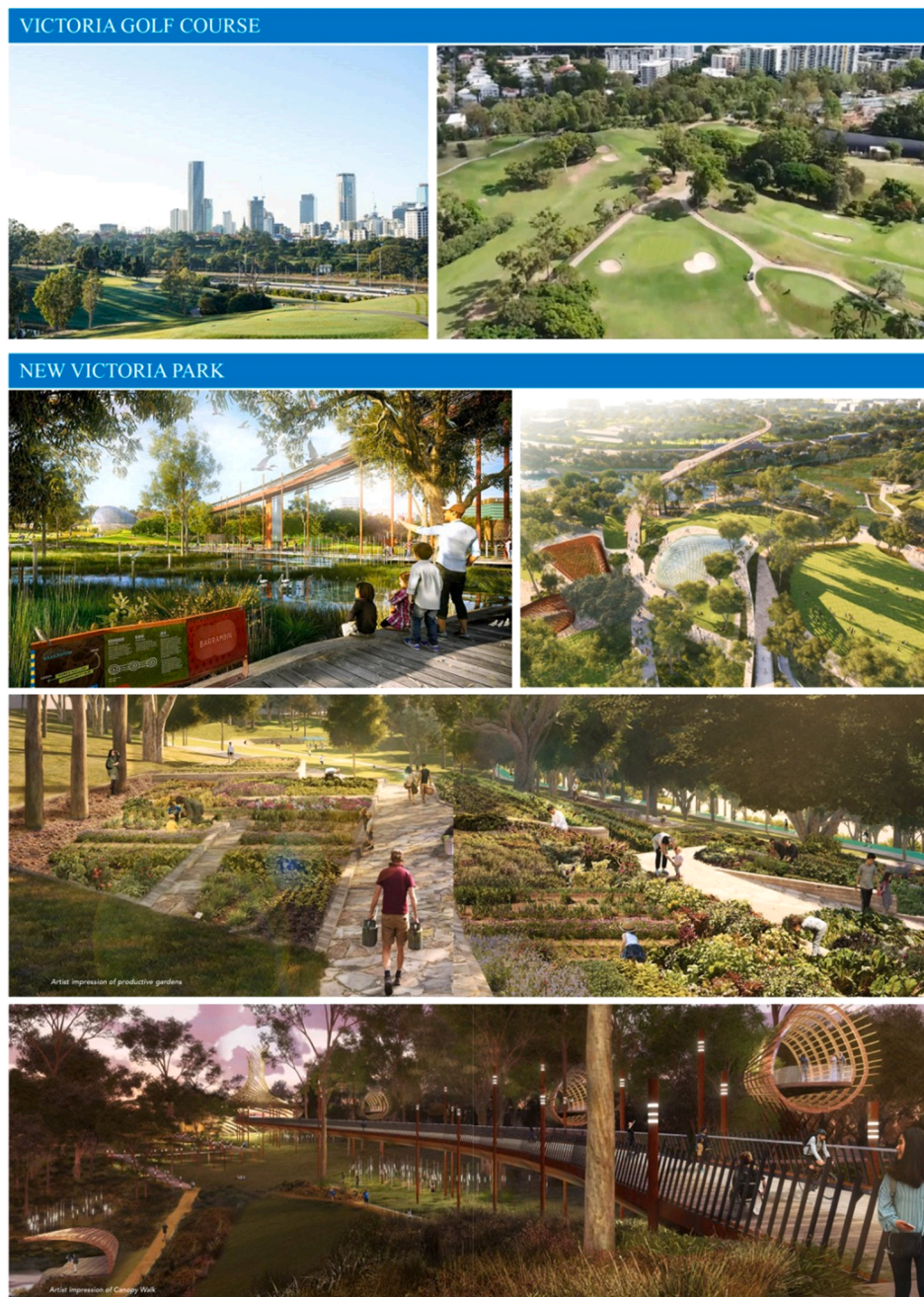


Fig. 6. Vision for the Victoria Park as emerging from the consultation process (source: elaboration from Brisbane City Council, 2020). (source: elaboration from [Brisbane City Council, 2020](#)).

increase in housing property values and the possible negative impacts related to green gentrification effects.

There are several opportunities for expanding the results of the present study.

First, it could be interesting to add more cases in the initial dataset of housing prices. This could lead to a refined analysis of homogeneous market segments.

Further development of the study could consider a better characterization of the urban parks included in the model, broadening the parks' typologies.

Moreover, future improvement of accuracy in the analysis would entail measuring actual travel distance by various modes of transport (e. g., walking, bike riding, public transport, etc.)—and comparing distance and time to get to the park. For simplicity, the actual analysis only

considered the Euclidean distance from housings and different types of parks.

Additionally, the analysis could be further expanded including other variables which affect the amenities of urban parks, such as the view on the park. In this context, different authors highlighted the existence of a positive relationship of view of green areas to prices ([Luttik, 2000](#); [Hiebert and Allen, 2019](#); [Liao et al. 2022](#)). The consideration of this latter variable could reveal more precisely differences between those properties close to urban parks and with a view on that and those without this last characteristic.

It would also be interesting to add further data detailing the neighbourhood context into the analysis, considering socio-economic parameters such as income, rent, population, etc. with specific reference to the case of the Victoria Park redevelopment ([Lieske et al., 2019](#)).

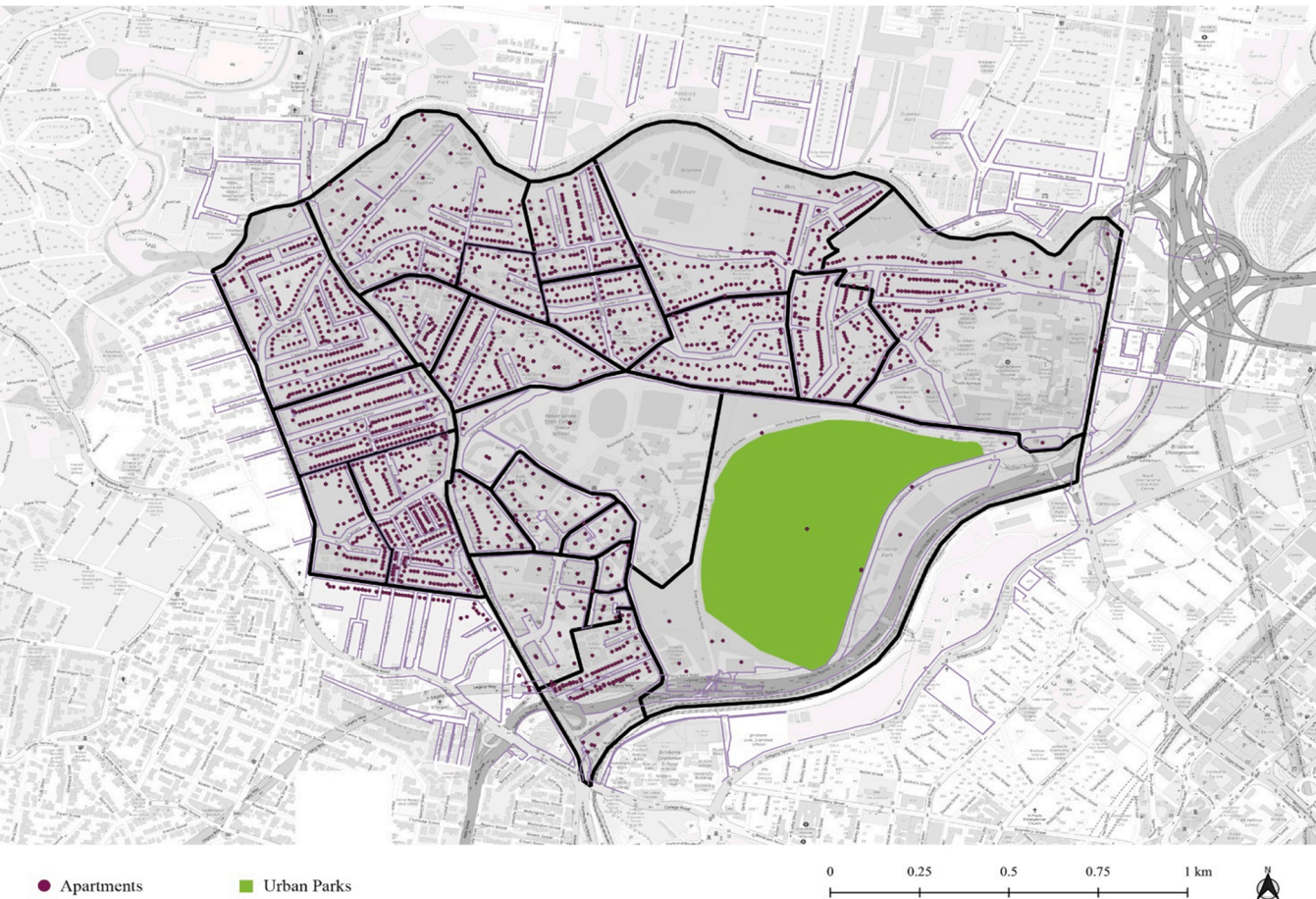


Fig. 7. The Victoria Golf Course Park and the distribution of the apartment sales data used in the analysis in the considered area.

Table 9
Synthetic data and results for the Kelvin-Grove and Herston districts.

Variables	Measure	Unit	Value
Average housing price	in the 750 m radius area	[AUS \$]	614,187
Dwellings	in the 750 m radius area	[number]	3430
Uplift	in the housing price due to the new park	[percentage]	3
Total uplift in the 750 m area		[AUS \$]	63,199,842.3

Finally, the results of the estimation could be used in the agent-based model simulation (Torrens and Nara, 2007) and participatory urban informatics workshops (Foth, 2018) for exploring the effect of the new park in terms of green gentrification.

CRediT authorship contribution statement

Marta Bottero: Conceptualization, Methodology, Formal analysis, Investigation, Data curation. **Caterina Caprioli:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation. **Marcus Foth:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation. **Peta Mitchell:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation. **Markus Rittenbruch:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation. **Marco Santangelo:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ufug.2022.127618.

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